A REPORT

ON

DESIGN, SIMULATION AND PERFORMANCE ANALYSIS OF 2-STAGE OPERATIONAL AMPLIFIER FOR THERMAL SENSOR APPLICATIONS.

BY

Shivakshit Patri, **2012AAPS168H**

Electronics and Communication

Prepared in the partial fulfillment of the
Practice School II Course

AT

Intel India Technology Private Limited

Intel Custom Foundry
Leading at the Edge of Moore's Law

A Practice School II Station of

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Acknowledgements

The internship opportunity I have with Intel Corporation is a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me though this period.

Bearing in mind previous I am using this opportunity to express my deepest gratitude and special thanks to Mr. Andy Bryant, Chairman of Intel Corp., Brian Krzanich, CEO and Kumud Srinivasan, President, Intel India who have given me this opportunity of pursuing my internship at Intel.

I express my deepest thanks to Ms. Swapna Kulkarni, Lecturer, BITS Pilani for taking part in useful decisions & giving necessary advices and guidance for an effective internship experience.
I choose this moment to acknowledge her contribution gratefully.

It is my radiant sentiment to place on record my best regards, deepest sense of gratitude to
Mr. Parveen Kumar, Manager, ICF AIP BA and Mr. Manish Srivastav, Design Engineer, ICF AIP BA for their careful and precious guidance which were extremely valuable for my study and work both theoretically and practically. It is due to their constant mentorship and efforts that I am now successfully able to contribute towards the projects of my team.

I perceive as this opportunity as a big milestone in my career development. I will strive to use gained skills and knowledge in the best possible way, and I will continue to work on their improvement, in order to attain desired career objectives.

Abstract

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
PILANI (RAJASTHAN)

Practice School Division

Station: Intel India Technology Private Limited **Centre:** Intel Custom Foundry, Bangalore

Duration: 6 Months **Date of Start:** 8th July, 2015

Date of Submission: 1st December

Title of the Project: DESIGN, SIMULATION AND PERFORMANCE ANALYSIS OF 2-STAGE OPERATIONAL AMPLIFIER FOR THERMAL SENSOR APPLICATIONS.

ID No. : 2012AAPS168H

Name: Shivakshit Patri

Discipline: B.E (Hons.), Electronics and Communication

Mentor's name: Mr. Parveen Kumar

Mentor's designation: Manager, ICF AIP BA

PS Faculty: Ms. Swapna Kulkarni, Lecturer, BITS Pilani

Key Words: Analog VLSI Design, Schematic Design, Operational Amplifier.

Project Areas: Circuit Schematic design of Op-Amp in 14 nm technology.

Abstract: The Objective of the project is to design a high gain, thermally stable 2-stage operational amplifier in 14nm Technology which can be used in the ADC for thermal sensor applications. The desired operational thermal range is -40^{o} C to 125^{o} C which is also known as the military thermal range. The design is also desired to work at extreme process variation (Slow, Slow and Fast, Fast) and voltage corners (5%).

Signature of Student
Shivakshit Patri

Signature of PS Faculty

Date: September 20th, 2015

Date:

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (RAJASTHAN)

Response Option Sheet

Station: Intel India Technology Private Limited Center: Intel Custom Foundry, Bangalore

ID No. & Name: 2012AAPS168H Shivakshit Patri

Title of the Project: **DESIGN, SIMULATION AND PERFORMANCE ANALYSIS OF 2-STAGE OPERATIONAL AMPLIFIER FOR THERMAL SENSOR APPLICATIONS.**

Usefulness of the project to the on-campus courses of study in various disciplines. Project should be scrutinized keeping in view the following response options. Write Course No. and Course Name against the option under which the project comes.

Refer Bulletin for Course No. and Course Name.

Code No.	Response Option	Course No.(s) & Name
1.	A new course can be designed out of this project.	Yes
2.	The project can help modification of the course content of some of the existing Courses	**EEE F313/ INSTR F313: ANALOG AND DIGITAL VLSI DESIGN**
3.	The project can be used directly in some of the existing Compulsory Discipline Courses (CDC)/ Discipline Courses Other than Compulsory (DCOC)/ Emerging Area (EA), etc. Courses	Yes
4.	The project can be used in preparatory courses like Analysis and Application Oriented Courses (AAOC)/ Engineering Science (ES)/ Technical Art (TA) and Core Courses.	No
5.	This project cannot come under any of the above mentioned options as it relates to the professional work of the host organization.	No

Signature of Student
Shivakshit Patri

Signature of Faculty

Date: 20th September, 2015

Date:

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1. Introduction

1.1 The Organization

Intel founded in Mountain View, California in 1968 by Gordon E. Moore and Robert Noyce, is a portmanteau of Integrated Electronics. Intel makes motherboard chipsets, network interface controllers and integrated circuits, flash memory, graphic chips, embedded processors and other devices related to communications and computing. Intel combines advanced chip design capability with a leading-edge manufacturing capability. Intel was an early developer of SRAM and DRAM memory chips, and this represented the majority of its business until 1981. Although Intel created the world's first commercial microprocessor chip in 1971, it was not until the success of the personal computer (PC) that this became its primary business. Recently, Intel has introduced a 3-D transistor that improves performance and energy efficiency. Intel has begun mass-producing this 3-D transistor, named the Tri-Gate transistor, with their 22 nm process, which is currently used in their 3rd generation core processors initially released on April 29, 2012. In June 2013, Intel unveiled its fourth generation of Intel Core processors (Haswell).

1.2 Intel Custom Foundry

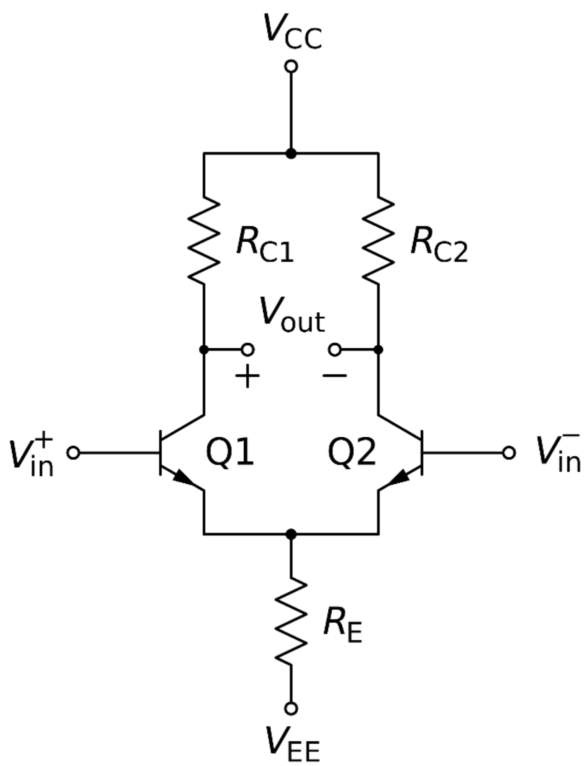
Founded in 2009, the Intel Custom Foundry division of Intel is the world's most advanced and complete contract IC manufacturer. It leverages the process development leadership, manufacturing prowess, and packaging resources of the world largest semiconductor company, Intel, to benefit its customers. With the help of advanced processes, wafer fabs, and packages, along with fully integrated assembly and test facilities, Intel supplies its foundry manufacturing capabilities to the customers in need. The Intel Custom Foundry (ICF) supplies foundry development kits (FDKs) and Wafers as final deliverables to its customers. ICF comprises of a number of teams namely, FDK, AIP, Memory and ASIC involved at various levels of manufacturing process.

2. Background Overview:

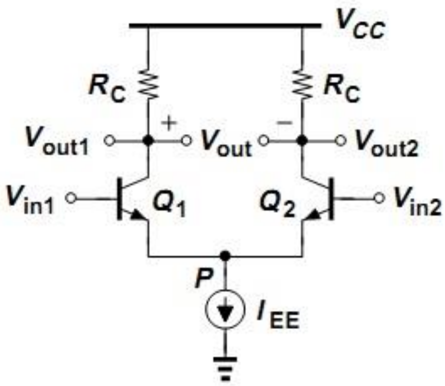
An **operational amplifier** ("op-amp") is a DC-coupled high-gain electronic voltage amplifier with a differential input and, usually, a single-ended output. In this configuration, an op-amp produces an output potential (relative to circuit ground) that is typically hundreds of thousands of times larger than the potential difference between its input terminals.

2.1 Operational Amplifier (Various Designs and Architectures):

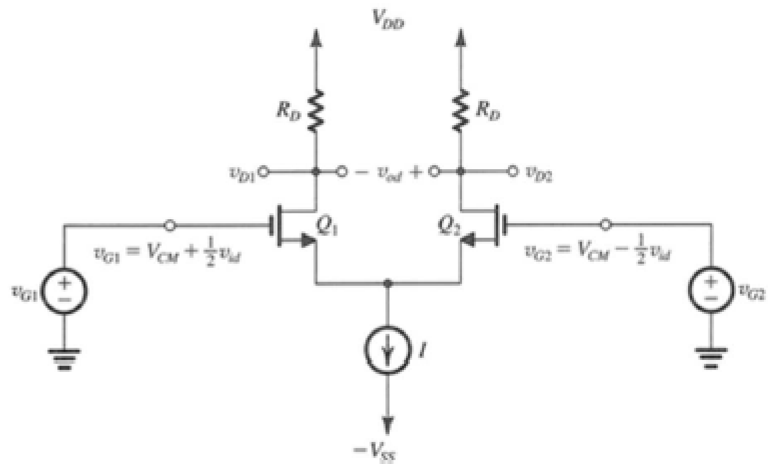
The simplest Operational Amplifier is made up of a single stage differential amplifier. It contains two common source amplifiers connected in parallel where the differential input is given to the gates and the output is obtained as a differential output across the load resistances.



But this design is not acceptable because the gain across the op-amp is dependent on the Bias input voltage and variation in the bias voltage causes huge distortion in the output. Hence to overcome that problem a current source is used instead of R_E .This Bias Current ensures that the Bias voltage at the input end remains constant.



Small Signal Analysis of an Op-Amp:



In the Operational Amplifier model suggested above has a small signal differential gain is given by

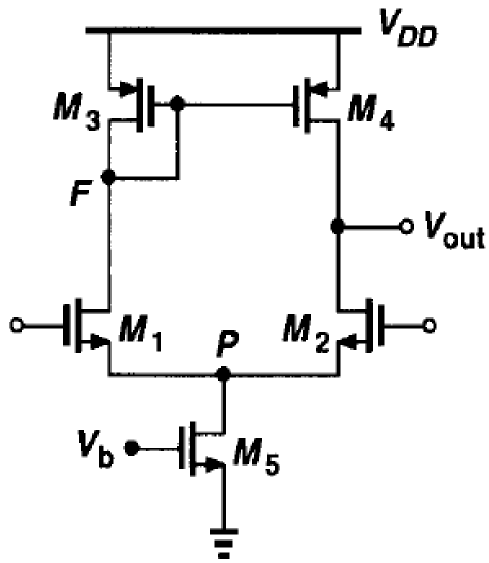
Differential gain:

$$A_d \equiv \frac{v_{od}}{v_{id}} = \frac{v_{o2} - v_{o1}}{v_{id}} = g_m R_D$$

We can infer from this Equation that the more is the load resistance the more will be the gain. This gives us the idea to use a current source at the load as the resistance of the current source is infinite.

And on further analysis we see that to have a single ended output instead of differential output a current-mirror active load seems more legit.

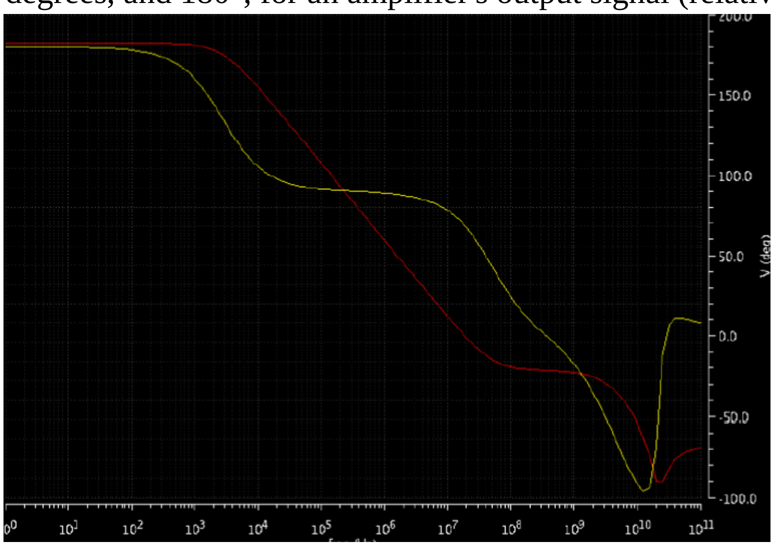
As the current source can be put on the layout, the current mirror is created using PMOS. Here is the circuit diagram of the Active Load Differential Pair (1-stage Op-Amp) –



The gain achieved in a single stage differential pair (around 35 dB) is often not sufficient for the practical applications. Therefore an additional stage is concatenated at the end of the differential pair to increase the gain. This stage contains the most simple common source amplifier.

Phase Margin:

In electronic amplifiers, the **phase margin** (PM) is the difference between the phase, measured in degrees, and 180°, for an amplifier's output signal (relative to its input), as a function of frequency.



Since an additional stage is added we find two poles in the bode plot of the Gain of the operational Amplifier.

The red line is the gain and the yellow line is the phase difference between input and the output. Phase margin is defined as the difference between 180 and the zero crossing of the phase plot. That is 180^o – (Zero Crossing). At poles we see 90^o decrement of phase. Therefore if the 2^{nd} pole occurs before 0 crossing then the Phase margin will be less than 0.Therefore to achieve 60^{o} phase margin we deliberately shift the poles. The gradient we see between pole1 to pole2 is constant and equal to -20dB/decade. Therefore we intentionally position pole1 at a lesser frequency to get the zero crossing before the 2^{nd} pole.

We also know that the frequency at which we have our first pole inversely depends on the load capacitance. Hence we try to vary the load capacitance on the first stage of our opamp.

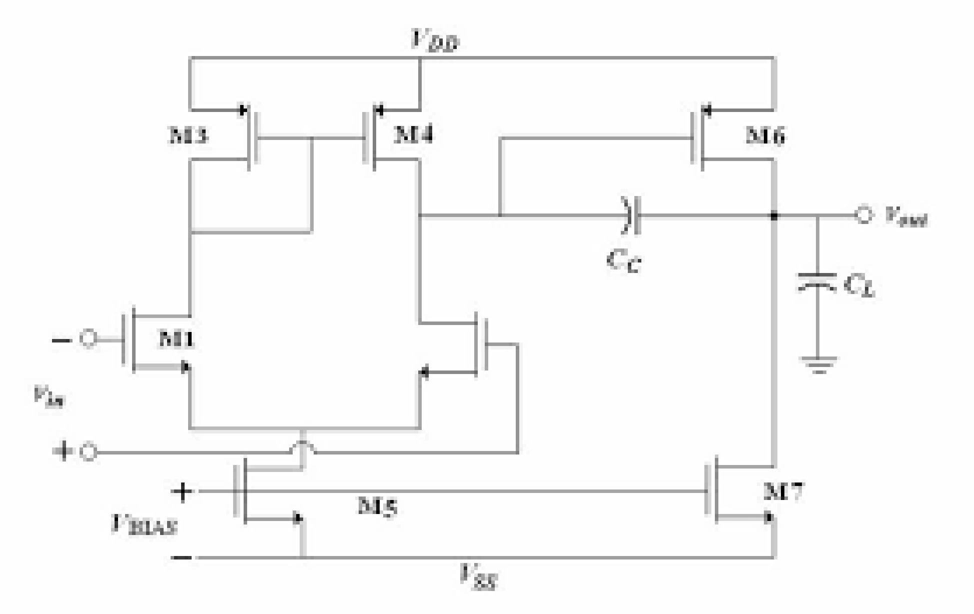
But for that we have to increase our capacitance by a very large amount. Instead we make use of the miller theorem to achieve this feat.

2.2 Miller Theorem:

A Circuit with feedback Capacitance equal to C is same as an open loop Circuit with input impedance equal to C(1+A) and output capacitance C(1+1/A) where A is the gain across the circuit.

We use this technique to increase the load capacitance on the first stage of our opamp which results in the shift of the first pole. Therefore we add a capacitor (Miller capacitance) as a feedback (between final output and the gate of the 2^{nd} stage NMOS) to the 2^{nd} stage to shift the pole first stage (1^{st} pole).

After this modification our circuit model looks like this:



3.1 A Brief Description about the Specifications (aimed) of the Operational Amplifier.

1. Gain : 65dB
2. V_{dd} : 1.8V
3. Unity Gain Bandwidth (UGB) : 5 MHz
4. Phase Margin : 60^o
5. Slew Rate : 5V/µsec
6. ICMR+ : 1.5V
7. ICMR - : 0.8V
8. Operational Temperature Range: -40^o to 125^o C.(Military Range)
9. Process Corners: Slow, Slow and Fast, Fast.
10. Voltage Corners: 1.71V, 1.80V and 1.89V (for our application 1.89 is sufficient).
11. Power : < 350µW
12. Load Capacitance : 15pF
13. PSRR : 60 – 100 dB
14. CMRR : 60 – 100dB

3.2 Design

Miller Capacitance:

Miller Capacitance = 0.25 x Load Capacitance = 4pF

Current Mirror current:

I_d = Slew Rate x Miller Capacitance = Slew Rate x Load Capacitance /4 = 20µA.

Sizing of the Mosfets:

There are 7 Mosfets in the schematic + 1 Mosfet to mirror the current in M5 and M7.Let's call it M8.

The sizing M1 and M2 depends on Unity Gain Bandwidth.
The sizing can be varied by changing the W/L ratio or the multiplication factor or the fingering factor f.
The sizing of M1 and M2 is given by

$$g_m = UGB \times C_c \times 2\pi = 250\ \mu A/V^2$$

The required g_m is achieved at M1 = 10.
While sizing M3 and M4 we should make sure that these mosfets doesn't go out of saturation at all times. The inequation that makes sure that the MOSFET is always under saturation is given by

$$M3 = M4 \leq \frac{2Id}{\mu nCox(Vdd - (ICMR+) - Vthp + Vthp)}$$

But there is no lower limit on the sizing factor. Having low sizing factor for M3, M4 makes sure that the resistance is high and hence the gain will be high.
We take the value M3 = M4 = 2.
Sizing of M5 the approach is similar and we find that the sizing here is M5 = 2.Sizing of M8 should be equal to M5 as it has to carry same amount of current as in M5.

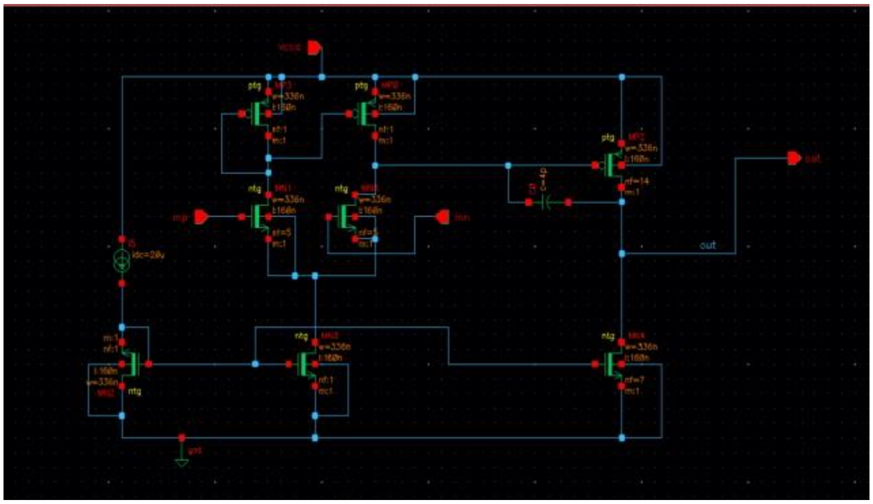
Sizing of M6:
It is desirable of the 2nd Pole arrives a decade after the first pole for maximum Unity Gain Bandwidth. For this $g_{m6} \geq 10 \times g_{m4}$ .This happens when M6 = 20.
M7 is the extension of the current mirror network of M5 and M8.It has to carry current that is delivered by M6 and should satisfy the equation M7/M8 = I7/I8.That happens at M7 = 10.

4.1 Performance Metrics:

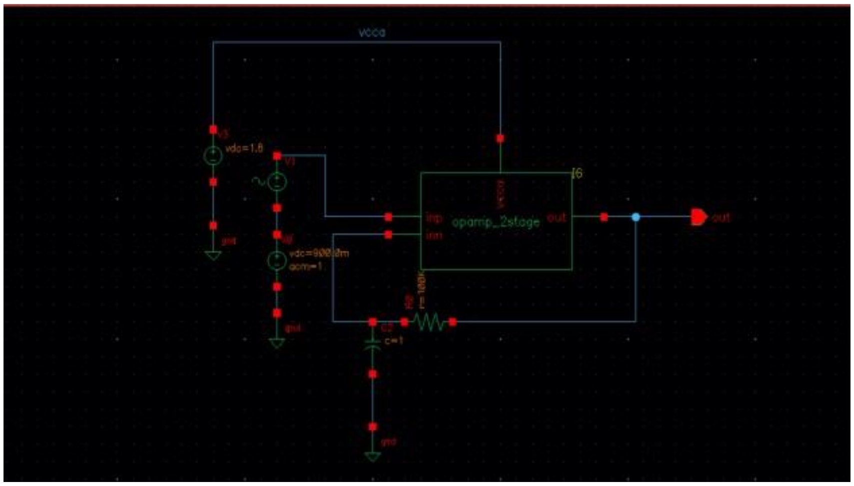
The Gain, UGB, Phase Margin etc. are calculated by using the feedback circuit:

Performance Measure of 14nm Two Stage Operational Amplifier Circuit:

Op-Amp Circuit

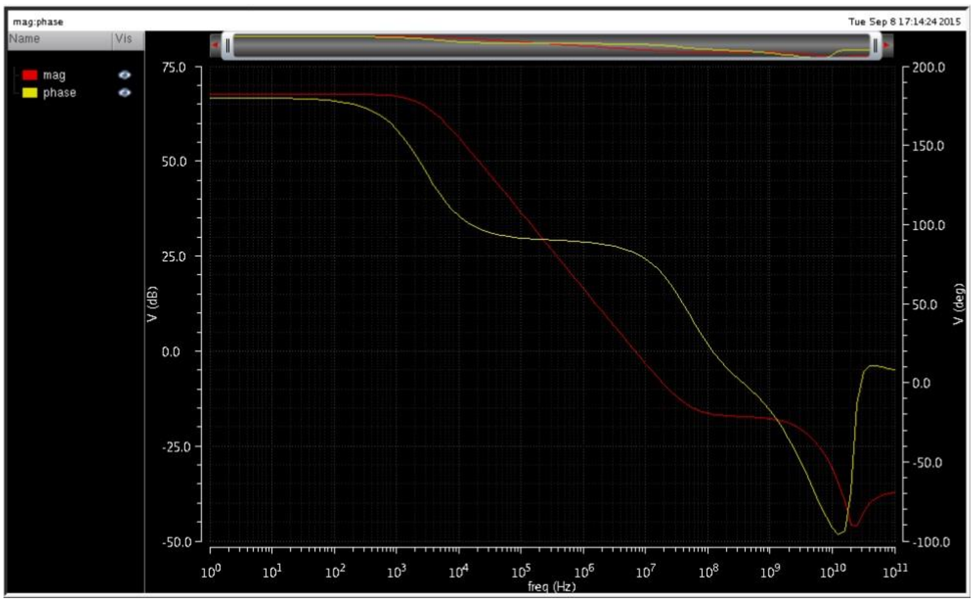


Test Circuit Used:



Results after Simulation:

Gain and Phase Plot:

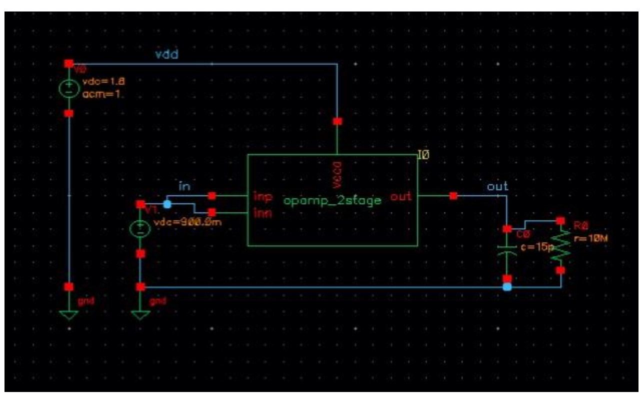


Results at Typical Case:

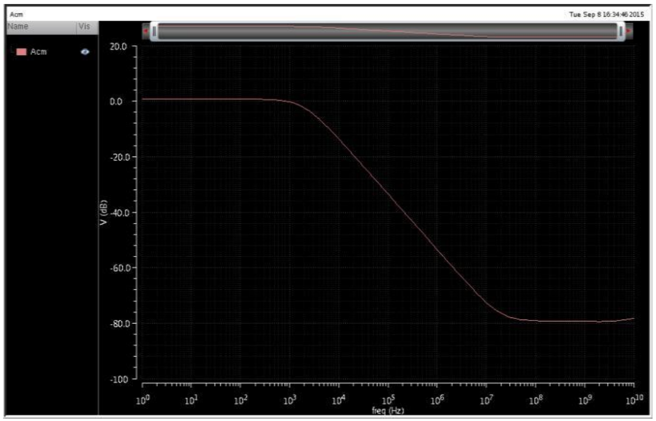
Plot Analysis:

Common Mode Rejection Ratio:

Circuit Used:

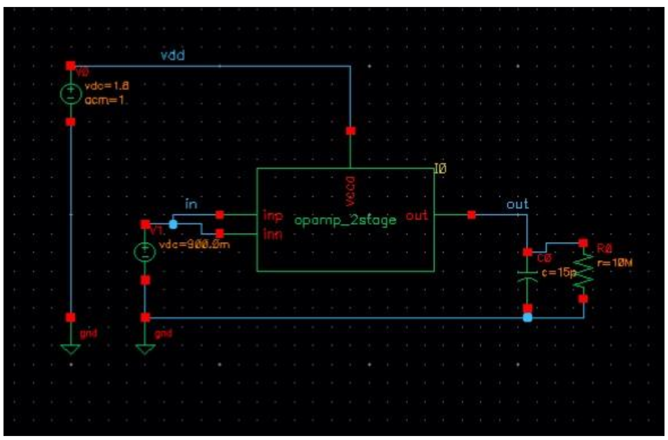


Common Mode Gain Plot:

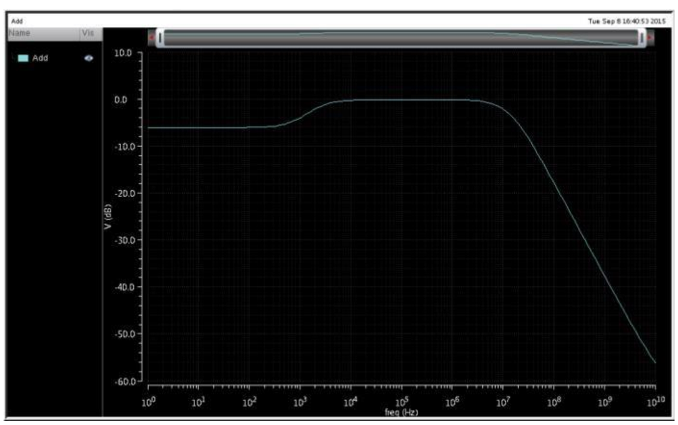


Common Mode Gain at lower Frequency is 0.7dB.
Therefore Common Mode Rejection Ratio = A_d / A_{cm} = 67.7dB – 0.7dB = 67 dB.
PSRR:

Circuit Used:



Power Supply Gain Plot:



Power Supply Rejection Ratio = A_d/A_{dd} = 67.7 – (-6.03) = 73.73 dB.

Power Consumption:

Current Drawn from the Power Supply Rails = 183.2uA.
Voltage at the Power Supply Rail = 1.8 V. Power Consumed = 1.8 x 183.2uA = 330uW (approx.)

Corner Voltages and Process Variations:

Assuming 5% Variation in the Supply Voltage.

At V_{dd} = 1.8 V

Temperature	Gain	UGB	Phase Margin
27$_o$C	67.61	6.81 MHz	82^o
53.98^{o}C*	67.00	6.40 MHz	82.2^o
-24.78^{o}C*	68.65	7.64 MHz	81.6^o

Operational Range = 54^o – (-25^o) = 79^o C

*The op-amp performance measures are observed to be drastically lesser than reasonable at the temperatures beyond the ones shown above.

At V_{dd} = 1.71 V

Temperature	Gain	UGB	Phase Margin
27$_o$C	67.15	6.80 MHz	82^o
77.84^{o}C*	65.93	6.14 MHz	82.3^o
6.33^{o}C*	67.60	7.13 MHz	81.8^o

Operational Range = 71.51^{o}C

At V_{dd} = 1.89 V

Temperature	Gain	UGB	Phase Margin
27$_o$C	67.97	6.83 MHz	82^o
30.35^{o}C	67.90	6.78 MHz	82^o
-56.70^{o}C	69.57	8.32 MHz	81.3^o

Operational Range = 87.05^{o}C

But generally the op-amp is supposed to work in the range -40^{o}C to 125^{o}C.

5.1 Tasks completed till Date:

1. Understood the file system in Linux and some important Linux Commands.
2. Learn how to design, simulate and Analyze Analog circuits in Cadence Virtuoso.
3. Learned the Design and Performance Metrics of Op-Amp.
4. Learned the working of Delta-Sigma ADC.

5.2 Future Tasks:

1. Make the Opamp work by tuning the sizing of the FETS in the corners at -40^{o}C to 125^{o}C.
2. Design and Simulation of Delta-Sigma ADC for Thermal Sensor.

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